Task 1: Student CGPA Calculator (C++)

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Objective

Build a C++ program that takes subject names, credits and grade points for a student and calculates the CGPA using the formula: CGPA = Σ(credit × grade point) / Σ(credits).

Approach

The program uses two classes. **Subject** stores the subject name, credit and grade point. **Student** stores the student name and a vector of subjects, with methods addSubject(), calculateCGPA() and displayCGPA(). main() reads the input and displays the final CGPA.

Source Code

```
#include <iostream>
#include <vector>
#include <string>
#include <iomanip>
using namespace std;

class Subject {
public:
    string subjectName;
    int credit;
    float gradePoint;

    // constructor
    Subject(string name, int cr, float gp) {
        subjectName = name;
        credit = cr;
        gradePoint = gp;
    }
};

class Student {
public:
    string name;
    vector<Subject> subjects;

    // constructor
    Student(string studentName) {
        name = studentName;
    }

    void addSubject(string subjectName, int credit, float gradePoint) {
        Subject newSubject(subjectName, credit, gradePoint);
        subjects.push_back(newSubject);
    }

    float calculateCGPA() {
        int totalCredit = 0;
        float weightGradePoint = 0;
        for (const Subject &subject : subjects) {
            totalCredit += subject.credit;
            weightGradePoint += subject.credit * subject.gradePoint;
        }
        return (totalCredit > 0) ? weightGradePoint / totalCredit : 0;
    }

    void displayCGPA() {
        cout << "\nStudent Name: " << name << endl;
        cout << fixed << setprecision(2);
        cout << "CGPA: " << calculateCGPA() << endl;
```

```
    }
};

int main() {
    string studentName;
    int numSubjects;

    cout << "Enter student name: ";
    getline(cin, studentName);

    Student student(studentName);

    cout << "Enter the number of subjects: ";
    cin >> numSubjects;
    cin.ignore(); // ignore the newline character

    for (int i = 0; i < numSubjects; i++) {
        string subjectName;
        int credit;
        float gradePoint;

        cout << "Enter subject name: ";
        getline(cin, subjectName);

        cout << "Enter credits for " << subjectName << ": ";
        cin >> credit;

        cout << "Enter grade points for " << subjectName << ": ";
        cin >> gradePoint;
        cin.ignore(); // ignore the newline character

        // add the subject to the student's record
        student.addSubject(subjectName, credit, gradePoint);
    }

    // calculate and display the CGPA
    student.displayCGPA();

    return 0;
}
```

Sample Output

```
Enter student name: Harshada
Enter the number of subjects: 3
Enter subject name: Maths
Enter credits for Maths: 4
Enter grade points for Maths: 9
Enter subject name: C++ Programming
Enter credits for C++ Programming: 4
Enter grade points for C++ Programming: 8.5
Enter subject name: Data Structures
Enter credits for Data Structures: 3
Enter grade points for Data Structures: 8

Student Name: Harshada
CGPA: 8.55
```